MIL-B-43706A
20 March 1980
MIL-B-43706
10 June 1970

MILITARY SPECIFICATION

BAGS, PLASTIC, BREAD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers plastic bags to be used in the packaging of bread by automatic packaging machines.

1.2 Classification. Plastic bread bags covered by this specification shall be of one type and two sizes, as specified (see 6.2).

Size A - 7-3/4 by 20 by 4 inch (197 by 508 by 102 mm) wide bottom (when open), see figure 1.
Size B - 8-1/4 by 15-5/8 by 4-1/8 inch (210 by 397 by 105 mm) wide bottom (when open), see figure 1.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8105

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SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge Polyolefin
UU-P-268	-	Paper, Kraft, Wrapping
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-B-43666	-	Boxes, Shipping, Consolidation

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletization and Containerized Unit Loads 40" by 48" Pallets, Skids, Runners, or Pallet-Type Base

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

US DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies shall be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

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3. REQUIREMENTS

3.1 Material. The material from which bread bags shall be fabricated shall conform to type I, class 2, grade B or C, finish optional, polyethylene of L-P-378, having a thickness of not less than 0.001 inch (0.03 mm), and shall conform to Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder.

3.2 Construction.

3.2.1 Bags. Bags shall be made from one or two pieces of polyethylene and shall conform to the construction and dimensions specified for size A or B bag in figure 1, as applicable.

* 3.2.2 Bag lip. The bags shall have a 1-1/2 inch (38 mm) long lip on one face at the open end of the bag (see figure 1). The lip shall have two 3/8 inch (10 mm) diameter holes for insertion on wicket arms. The holes shall be positioned as shown in figure 1. One-quarter inch (6 mm) slits shall be cut on the lip above the center line of the wicket holes as shown in figure 1.

* 3.2.3 Bag seams. Bag seams shall be continuously sealed along their entire length. The location and width of the seams shall be as shown in figure 1. The seam strength shall be such that the seams shall show no evidence of separation when tested as specified in 4.3.1. Leakage from not more than six pinholes in the seams of a single bag shall not be construed as seam separation.

3.2.4 Bottom gusset. The bag shall have a bottom gusset. The gusset ends shall be heat sealed with the side seams of the bag (see figure 1) to permit the end of the filled bag to conform to the shape of the end of the loaf of bread. The gussets shall be of widths as specified in figure 1.

3.2.5 Wickets. Metal wickets shall be those commercially used with plastic bread bags, with two arms that will fit the two 3/8 inch (10 mm) diameter holes in the bag flap (see figure 1), of such a length as to hold 750 bags.

3.3 Blocking. Bags shall show no signs of blocking when tested as specified in 4.2.2.2.

3.4 Workmanship. The bags shall have clean cut edges, and the surface shall be free of wrinkles, tears, holes or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document. The sample unit shall be one bag. The lot size shall be expressed in units of one bag.

4.2.2 End item inspection. The sample unit shall be one bag. The lot size shall be expressed in units of one bag.

4.2.2.1 End item examination.

* 4.2.2.1.1 Visual examination. The end item shall be examined for the defects listed in table I. The inspection level shall be I and the acceptable quality level (AQL) shall be 4.0 defects per hundred units.

TABLE I. Examination of workmanship and construction

Examine	Defect
Workmanship	Tear, cut, split or hole. Surface not clean, wrinkled, scuffed.
Construction	Lip missing; holes in lip missing, lip slits missing or not above center line of holes. Seams not flush with outer edge of bags. Bottom of bags not gusseted. Edges of gussets not sealed with the side seal of the bags.

* 4.2.2.1.2 Dimensional examination. The end item shall be examined for the defects listed in table II. The inspection level shall be S-2 and the AQL shall be 4.0 expressed in defects per hundred units.

TABLE II. Examination of dimensional defects

Examine	Defect
Bag	Does not conform to dimensions shown on figure 1 for size A or B bag, as applicable. Thickness not as specified.
Side seam width	Greater than 1/32 inch (1 mm) in width.

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* 4.2.2.2 End item testing. The end item shall be tested for the characteristic indicated in table III. The sample unit shall be one bag. The lot size shall be expressed in units of one bag. The lot shall be unacceptable if one or more units fails to meet any requirements specified. Test results are applicable to unit requirements.

TABLE III. Instructions for testing end item

Characteristic	Rqmt para	Test method	Number of determinations per sample unit	Results reported as pass or fail 1/
Seam strength	3.2.3	4.3.1	One	X
Blocking	3.3	4.3.2	One	X

1/ If failure is indicated, report description thereof.

* 4.2.3 Packaging inspection. An examination shall be made to determine that packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with table IV. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Defects of closure shall be examined on shipping containers fully packaged. The lot shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

TABLE IV. Examination of packaging

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Bulged or distorted container.
Contents	Number per wicket is more than five bags under count. 1/ Less than four wickets per container.

1/ Count one wicket per sample container.

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* 4.2.3.1 Examination for palletization. When applicable, examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored as specified in table V. The lot size shall be the number of palletized unit loads in the end item inspection lot. The sample unit shall be one palletized load. The inspection level shall be S-1 and the AQL shall be 6.5 expressed in defects per hundred units.

TABLE V. Examination for palletization

Examine	Defect
Finished dimension	Length, width or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.2.3.2 Examination of consolidation boxes. When applicable, consolidation boxes shall be examined for assembly, closure and reinforcement in accordance with the appendix of MIL-B-43666.

4.3 Methods of inspection.

4.3.1 Seam test. The bag shall be not less than 1/2 filled with water and suspended not less than 5 minutes. After the suspension period, the portion of the seams in contact with the water shall be examined for seam separation. Any nonconformance to the requirements of 3.2.3 shall constitute failure and shall be cause for rejection of the lot.

4.3.2 Blocking test. Two, 4 by 4 inch (102 by 102 mm) specimens shall be cut from a bag. The specimens shall be placed together with the outer bag surfaces in contact with each other. The stacked specimens shall be placed between two glass plates and placed in an oven at a temperature of 150 $\pm$ 2°F (66 $\pm$ 1°C). A 4 pound (1.8 kg) weight shall be placed on top of the glass plates. After not less than 48 hours, the specimens shall be taken from the oven and from between the glass plates and be allowed to cool for not less than 1 hour. The specimens shall then be slowly separated by hand and any evidence of blocking (sticking) shall be reported. In accordance with 3.3, any blocking shall constitute failure of the test and shall be cause for rejection of the lot.

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5. PACKAGING

5.1 Packing. Packing shall be of level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Seven hundred fifty bags shall be placed on a wicket commercially used by industry (see 3.2.5). Four loaded wickets shall be packed flat in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each container shall have the contents completely covered on the top and bottom with a sheet of polyethylene film having a minimum thickness of 0.001 inch (0.02 mm) conforming to type I, class 2, grade and finish optional, of L-P-378. Alternatively, the contents shall be completely covered on the top and bottom with a sheet of 30 pound (48 g/m^2) minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of the container specification.

5.1.2 Level B. Seven hundred and fifty bags shall be placed on a wicket commercially used by industry (see 3.2.5). Four loaded wickets shall be packed flat in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each container shall have the contents completely covered on the top and bottom with a sheet of polyethylene film having a minimum thickness of 0.001 inch (0.02 mm) conforming to type I, class 2, grade and finish optional, of L-P-378. Alternatively, the contents shall be completely covered on the top and bottom with a sheet of 30 pound (48 g/m^2) minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification.

5.1.2.1 Weather-resistant containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636, and closed in accordance with method III as specified in the appendix of the box specification.

5.1.3 Level C. Seven hundred and fifty bags shall be placed on a wicket commercially used by industry (see 3.2.5). Four loaded wickets shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Each container shall have the contents completely covered on the top and bottom with a sheet of polyethylene film having a minimum thickness of 0.001 inch (0.02 mm) conforming to type I, class 2, grade and finish optional, of L-P-378. Alternatively, the contents shall be completely covered on the top and bottom with a sheet of 30 pound (48 g/m^2) minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

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5.2 Containerization. When specified (see 6.2), bags packed as specified in 5.1 shall be consolidated in a container conforming to type I, II or III, style and size optional of MIL-B-43666.

5.3 Palletization. When specified (see 6.2), bags, packed as specified in 5.1, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract, shipping containers and when applicable, consolidation boxes and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The bags are intended for use in bagging loaves of bread by automatic filling machines.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size of bag (see 1.2 and 3.2.1).
(c) Quantity required.
(d) Applicable level of packing (see 5.1).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.1.2.1).
(f) When shipping containers of bags are to be containerized (see 5.2).
(g) When shipping containers of bags are to be palletized (see 5.3).

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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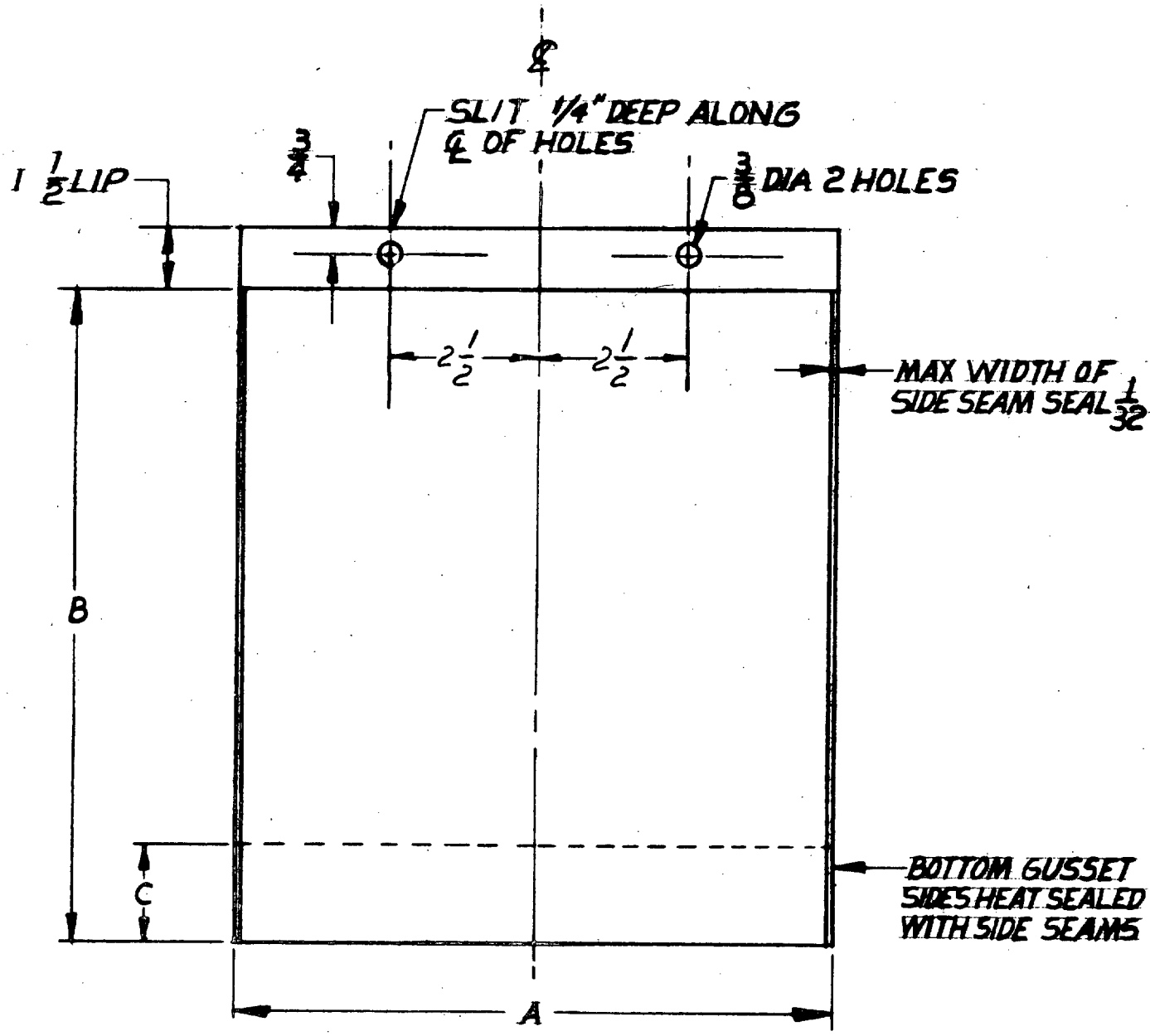
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SIZE A - A = 7 3/4" B = 20" C = 2" (4" WIDE WHEN OPENED)
SIZE B - A = 8 1/4" B = 15 5/8" C = 2 1/16" (4 1/8" WIDE WHEN OPENED)
TOLERANCES - ± 1/8" FOR DIMENSIONS 1" AND GREATER
± 1/16" FOR DIMENSIONS LESS THAN 1" EXCEPT 1/32" SEAM WIDTH

FIGURE 1 - BAGS, PLASTIC, BREAD

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